National University of Lesotho
Department of Physics and Electronics
EE3404: Classroom Questions and Assignment Questions

Classroom Questions

1. **Conceptual:** Why can capacitors and inductors be considered "memory" elements in circuits? How does this property affect circuit behavior over time?
2. **Application:** Your phone uses a power supply with capacitors inside. What role do these capacitors play in ensuring the phone runs smoothly?
3. **Conceptual:** If we double the area of a capacitor's plates while keeping distance constant, what happens to its capacitance and why?
4. **Application:** Inductors are widely used in car ignition systems. Why do you think an inductor's ability to store magnetic energy makes it suitable for this application?
5. **Problem-solving:** If an inductor has $L = 10mH$, and the current through it changes at a rate of $200A/s$, what is the voltage across it?
6. **Conceptual:** What is the difference between independent and dependent sources? Can you think of a real-world situation where a dependent source might appear?
7. **Application:** Radio tuning circuits rely on capacitors and inductors. Which properties of these elements make them useful for selecting specific frequencies?
8. **Problem-solving:** Two resistors $R_1 = 2k\Omega$ and $R_2 = 3k\Omega$ are in series with a 10V source. Use the voltage divider rule to calculate the voltage across each resistor.
9. **Conceptual:** Why must we "zero out" other independent sources when applying the principle of superposition?
10. **Challenge:** Imagine you have two independent voltage sources and one dependent source in a circuit. Would the superposition theorem still apply directly? Why or why not?

Assignment Questions

1. **Calculation (Capacitance Design):** A capacitor is constructed using two square metal plates, each of side length ***10 cm***, separated by ***1 mm*** of air. Calculate its capacitance. How would the capacitance change if the separation was halved?
2. **Derivation & Analysis (Inductor Energy):** Derive the energy stored in an inductor starting from $v = L\frac{di}{dt}$. Then calculate the energy stored when a 5 mH inductor carries 2 A of current.
3. **Circuit Analysis (Superposition with Real-World Application):** In a household circuit, assume you have a 12V DC supply and a 10V battery both connected with resistors. Using **superposition**, calculate the current in a 5Ω resistor. Clearly show each step.

4. **Design Challenge (Voltage Divider):** You are tasked with designing a simple voltage divider to step down a 12V car battery to 5V to power a small sensor. Choose appropriate resistor values and justify your choice, considering power dissipation and efficiency.
5. **Research/Conceptual (Real-World Application):** Dependent sources are widely used in transistor amplifier models. Write a short technical note (2–3 pages) explaining how dependent sources represent transistor behavior. Include diagrams and equations.